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Runner Version: 1.0.0
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This guide documents the internal architecture and design of the BriteTest framework and Runner API. It is companion document to the BriteTest Runner Internal Reference document.

Preface

This document is for contributors.

For a list of other BriteTest documents and the repository layout, see the BriteTest Documentation Guide (`BriteTest_Documentation_Guide.md`).

For a glossary of terms, see the BriteTest Glossary Reference (`BriteTest_Glossary_Reference.md`).

A printer-friendly PDF file for this document is available in `docs/pdf/`,

Document Version History

Document	Runner	Test	Date	Comment	Author/Editor
1.0	1.0.0	1.0.0	2026-06-11	Initial version.	Paul Sinclair

- The **Document** column records the document's version with the format `M.u` (Major, update).
- The **Runner** column records the BriteTest Runner API version current at the time this document version was published and is defined by its `BT_RUNNER_VERSION` macro.
- The **Test** column records the BriteTest Test API version current at the time this document version was published and is defined by its `BT_TEST_VERSION` macro.
- Both Runner and Test use the version format `"M.m.p"` (Major, minor, patch).
- `M` is the same for the Document, Runner, and Test versions.

The document's update version tracks released updates to this document and does not correspond to a minor or patch version. `u` increments when document changes are published in a release without a change to `M`, and it resets to `0` when `M` is incremented.

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1. Introduction

Basic concepts and high-level description of the framework internals and design goals.

In this document, a *symbol* refers to any named entity in the BriteTest framework, including:

- typedefs
- structs
- enums and enum values
- macros
- variables
- functions

BriteTest exposes some internal symbols in `britetest_runner.h` because the framework's macros expand into code that depends on internal types and helper functions. These symbols must be visible to user code for the macros to compile correctly, but they are not part of the public API and should not be used directly.

Their presence in the header reflects C implementation requirements, not intended usage. Contributors may modify these symbols as needed, provided the public API contract remains intact.

1.1 Public and Internal Naming Conventions

Any framework names that are public and visible to BriteTest API users are prefixed with `bt_...` (typically lowercase) or `BT_...` (typically uppercase).

Any framework names that are internal (but technically visible to BriteTest API users) follow the pattern `britetest_..._internal_t`, `britetest_..._internal`, or `BRITETEST_..._INTERNAL`. These names should not be referenced by API users.

In general, users of the API should not define names prefixed with `bt_`, `BT`, `britetest_`, or `BRITETEST_`, or reference names prefixed with `britetest_` or `BRITETEST_`.

2. Architecture Overview

- Process model
- Thread model
- Signal handling
- Fault detection
- Execution flow

3. Orchestrator Internals

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- Argument parsing
- Report lifecycle
- Category aggregation

4. Test Group Internals

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5. Test Execution Engine

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6. Process-Isolated Execution

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7. Thread-Isolated Execution

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8. Signal Guard System

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- Fault classification
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9. Report Generation Internals

- Output formatting
- Category totals
- Fault messages
- Notes and metadata

10. Internal State and Global Variables

(Placeholder for internal state descriptions.)

11. Error Handling and Safety Guarantees

(Placeholder for internal error semantics.)

12. Implementation Notes

- Portability considerations
- POSIX dependencies
- Platform differences

13. Future Improvements

(Placeholder for roadmap items.)

Glossary

For general BriteTest terms, see the BriteTest Glossary document.

Runner-Specific Terms:

- **Orchestrator Lifecycle**: The sequence of initialization, group execution, test execution, and report finalization performed by the BriteTest framework.
- **Guard Behavior**: The mechanism BriteTest uses to catch runtime faults (e.g., segmentation faults) and continue executing remaining tests.
- **Isolation Semantics**: The rules governing how tests and groups run in threads or processes to prevent interference and ensure fault containment.
- **Concurrency Model**: The framework's rules for running tests concurrently within `BT_BEGIN_CONCURRENT` / `BT_END_CONCURRENT` blocks.
- **Execution Phases**: The internal stages of orchestrator operation, including initialization, argument parsing, group dispatch, test dispatch, and report writing.
- **Fault Handling Model**: The framework's strategy for capturing and reporting faults without terminating the entire test run.
- **Control File Promotion**: The process of replacing an outdated control file with a newly generated file when differences are expected or intentional.
- **Nested Group Behavior**: The rules governing how groups may contain other groups and how isolation and concurrency propagate through nested structures.
- **Concurrent Block Behavior**: The semantics of executing multiple tests in parallel within a concurrent block, including ordering and isolation rules.